# Project Smart Layer by Azam Puri

What the mill needs to provide to install the app (read-only on your ERP)

## How it runs (read this first)

Smart Layer has two parts:

- **The engine** - reads the ERP, runs the dashboard, sends the alerts and does the automatic syncs. It is installed on **one always-on Windows PC** near the ERP.
- **The viewer** - the dashboard is just a web page. The owner and managers open it in **any browser** (Windows, **Mac**, iPhone, Android). Nothing to install, and their device does **not** need to stay on.

So the app is installed on the always-on PC only. The owner's MacBook, laptop or phone just opens the dashboard in a browser - on the mill network, or from anywhere through a secure Tailscale link. Do not run the engine on a personal laptop that sleeps or switches off, or it will miss syncs and alerts.

## 1. A computer to run it (the always-on PC)

- An always-on Windows PC (not the ERP database server itself). Windows 10/11 or Server, 64-bit.
- Modest spec is fine: 8 GB RAM, ~20 GB free disk. It only reads the ERP and serves a dashboard.
- It must be on the same network as the ERP database server.

## 2. ERP database connection (read-only)

- Which database the ERP uses: **Oracle**, SQL Server, MySQL or PostgreSQL, and its version.
- The database address: **host : port / service** (for Oracle, e.g. `192.168.18.199:1521/orcl`), or the TNS name.
- A **read-only** database user and password. Best practice: create a dedicated read-only user (the app prints the exact SQL with `SmartLayer.exe grants`). Never give a write/admin login.
- If payroll or other data lives in a **separate schema**, that read-only user must be allowed to read it too.

The password is typed once during setup and stored only on this PC (sealed to the machine on Windows). It is never shown again and never leaves the mill.

## 3. Oracle 11g only - the Oracle Client

- Older Oracle (11g and earlier) needs the free **Oracle Instant Client 19 or newer** (Windows x64) on the PC.
- Download Basic Light, unzip to e.g. `C:\oracle\instantclient_23`, and note that folder (it has `oci.dll`). Setup asks for it.
- This does not touch or change your existing Oracle install - it sits alongside, used only by this app.

## 4. Network access

- The PC must reach the ERP database server over the LAN (e.g. Oracle port 1521 open to this PC).
- To open the dashboard to phones/PCs on the mill network, the app opens its dashboard port (default 8080) in the Windows firewall during install.

## 5. Choices to make at setup

- A **dashboard password** of your choice (type it; don't leave it blank).
- Whether to open the dashboard to the whole mill network or keep it on this PC only.

- Optional - the owner's phone: install the free **ntfy** app for alerts (setup gives a private topic to subscribe to).
- Optional - a **Claude API key** if you want the 'Ask the ERP' chat.

## 6. Licence (one step)

- After install, run `SmartLayer.exe machine-id` and send us that ID.
- We send back a licence file/code, tied to that one PC. Paste it on the dashboard's Licence screen.

## 7. If the ERP is one we have not mapped before

- The app reads each ERP through a mapping file. We already have mappings for known ERPs.
- For a new ERP, we run a few read-only queries to list the relevant tables and columns, then write its mapping - a one-time step, after which everything works the same.

| The app NEVER | |
|---|---|
| **Writes to the ERP** | every connection is read-only (SET TRANSACTION READ ONLY) |
| **Sends mill data out** | all data stays on the mill's own PC |
| **Needs admin DB rights** | a plain read-only login is enough |